

Combining Map and Cross Section Results into a VisibleGeology model

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This exercise is designed to follow an earlier assignment in which students create a geologic map and cross section of a field area. For EMU's ESSC 355 Geological Field Methods course, the field area is a subset of the Argenta, MT 1:24,000 topographic map.

Introduction (with geoscience context)

Bedrock geologic maps and cross sections are each two-dimensional representations of three-dimensional geologic structures. As such, the information communicated in a map and cross section drafted for the same area must be internally consistent. This lab activity is designed to help students generate a three-dimensional representation of a geologic field area using data contained within a hand-drawn geologic map and cross section.

Learning Goals

- Strengthen spatial reasoning skills needed to synthesize maps and cross sections into three-dimensional models.
- Use the process of model construction to support interpretation of a geologic history.
- Predict the type and geometry of structures located outside of a field area based on three-dimensional model

Pre-activity assignments/reading (if applicable)

To complete this assignment, students must have already completed a final or near-final version of a map and cross section for their field area. Students are also encouraged to visit the Geology Explorer program within Visible Geology to get comfortable with its functionality prior to starting the activity.

****in progress** short homework used for introducing students to the following controls**

Adding and editing layers, making topography, faulting, folding according to specifications, tilting, eroding

Using the Visible Geology Tools

- To access Visible Geology, use the web address here: <https://www.visiblegeology.com/>
- Once you open the site, click the Geology Explorer pane to open a new model.
- The new model can be edited by adding layers of horizontal sedimentary rock as a starting condition, and then modifying those layers through a number of possible brittle and ductile deformation events, erosion, and intrusions. These modifiers are found in the "events" panel.
- A brief tutorial on functionality is provided here (view "How to build a 3D Geologic model"): <https://www.seequent.com/products-solutions/academic-program/visible-geology/support/>

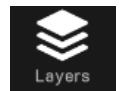
Activity (description and steps)

1. Summarize your map and cross section

Before generating a model using the Geology Explorer tool, describe the structure(s) represented in your map and cross section

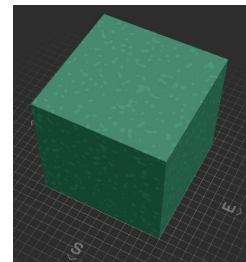
2. Set up your model

- Once you open your new model, navigate to the topography function on the left-side panel.
- Select the option to make a new topography, shown at the top of the menu.
- Keep the default settings in the east-west (blue arrow) and north-south (red arrow) directions; reduce the vertical dimension (green arrow) by approximately half.
- Use the various topography editing features to *roughly* mimic the topography of your **entire field map** (not just the area you have already mapped using field data and satellite imagery). Note that you can change the radius and strength of each editing feature to help with editing. Roughly mimicking topography means representing which regions of the map area are highest, which regions are lowest, and the position (trend, spacing) of prominent ridges. Do not spend too much time on this step... you can always return later to edit.
- Give your topography a name, then click “save and return”. Your default model should now appear with your specified dimensions and topography.
- Now, **add layers**. Add at least the number of layers that are present in your map legend. You can rename them, recolor them to match your legend, and change their thicknesses. Just try the default thickness to start and revisit later.
- Navigate to the “events” menu where you will find operations to fault, intrude, fold, tilt, and erode the rocks in your model. Your task now is to perform whatever series of operations is necessary to produce a 3-D model of your field area that matches your map and cross section. While you are doing this, you will likely need to edit your layer thicknesses and positions, and perhaps the topography as well.



Note: You will almost certainly not match your map and cross section perfectly. This is okay. Remember, you only roughly mimicked the topography, you have not calculated true unit thicknesses, some structures might be too small to model accurately in the Geology Explorer, and you may have made mistakes in your map and cross section. Focus on representing the gross structure of the field area.

- When you are happy with your model, paste a screenshot of it on the next page, taken from a southeastern view point as in the example at right.





Visible Geology

Activities that emphasize geoscience thinking competencies using the Visible Geology applications

3. *Investigate your model*

- a. What structure(s) did you introduce in your model? Be as specific as possible. For example, if your model has a fault, report the type of fault, its strike, dip, dip direction, the slip direction (reported as rake), and the slip magnitude. If you added a fold, report the type of fold, plunge direction and magnitude, vergence, approximate interlimb angle, etc. If you added an unconformity, state which units unconformably overlie others.

- b. Compare and contrast your model and your paper map and cross section. To do this, you will want to reproduce approximately the same line of section from your paper map by using the xSection tool. Where you note discrepancies, offer an explanation as to why they exist.



Possibly add a part asking students to make another cross-section through their model after they have validated the model with their own cross-section so they can see how the model allows them to quickly visualize structures from a different orientation and see how the structures look different.

- c. Based on the events you included in your model, develop a geologic history of your field area. For each major event, explain how you know its relative timing. Remember, deposition and lithification of units as well as erosion should be included in your geologic history.

3. *Use your model*

Your paper map and cross section were produced using data from only a restricted portion of your larger map area. Those data informed your model, which now (if instructions were followed) represents the geology in the entire map area. Based on your model, what structure(s) is/are present in the western portion of the larger map



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area? Be specific, referencing place names from your map area as needed. If you note major discrepancies between your model and satellite imagery, describe them.